



DOWLEX™ 2685G

Polyethylene Resin

Overview

DOWLEX™ 2685G Linear Low Polyethylene Resin is designed for the production of a wide variety of film applications. Films made from this resin exhibit a combination of good toughness and tear resistance.

Complies with:

- U.S. FDA FCN 741
- Canadian HPFB No Objection
- EU, No 10/2011

Consult the regulations for complete details.

Additive

- Antiblock: 5100 ppm
- Slip: 917 ppm
- Processing Aid: No

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	0.920 g/cm ³	0.920 g/cm ³	ASTM D792
Base Density ¹	0.917 g/cm ³	0.917 g/cm ³	Dow Method
Melt Index (190°C/2.16 kg)	0.75 g/10 min	0.75 g/10 min	ASTM D1238
Films	Nominal Value (English)	Nominal Value (SI)	Test Method
Film Thickness - Tested	1 mil	25 µm	
Film Puncture Energy	41.0 in·lb	4.63 J	
Film Puncture Force	14.0 lbf	62.3 N	
Film Puncture Resistance	246 ft·lb/in ³	20.4 J/cm ³	
Film Toughness			ASTM D882
MD	1610 ft·lb/in ³	133 J/cm ³	
TD	1600 ft·lb/in ³	132 J/cm ³	
Secant Modulus			ASTM D882
1% Secant, MD	23300 psi	160 MPa	
2% Secant, MD	19400 psi	133 MPa	
1% Secant, TD	26500 psi	183 MPa	
2% Secant, TD	21400 psi	147 MPa	
Tensile Strength			ASTM D882
MD : Yield	1630 psi	11.2 MPa	
TD : Yield	1620 psi	11.2 MPa	
MD : Break	8160 psi	56.3 MPa	
TD : Break	6870 psi	47.3 MPa	
Tensile Elongation			ASTM D882
MD : Break	610 %	610 %	
TD : Break	690 %	690 %	
Dart Drop Impact	630 g	630 g	ASTM D1709A
Elmendorf Tear Strength			ASTM D1922
MD	400 g	400 g	
TD	520 g	520 g	
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Vicat Softening Temperature	216 °F	102 °C	ASTM D1525
Melting Temperature (DSC)	246 °F	119 °C	Dow Method
Optical	Nominal Value (English)	Nominal Value (SI)	Test Method
Gloss (45°)	47	47	ASTM D2457
Haze	16.0 %	16.0 %	ASTM D1003

Extrusion Notes

Fabrication Conditions For Blown Film:

- Die Size: 8 in.
- Screw Type: DSB II
- Die Gap: 70 mil
- Melt Temperature: 437°F
- Output: 12 lb/hr/in. of die circumference
- Die Diameter: 8 in.
- Blow-Up Ratio: 2.5:1
- Screw Speed: 39 rpm
- Frost Line Height: 43 in.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Base density is estimated using the assumption that every 1000 ppm of antiblock in the finished product raises the density of the polymer by 0.0006 g/cm³. Base density is the estimated density of the polymer if it did not contain any antiblock.

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